SUSTAINABILITY AND MANAGEMENT

Topics

1. Need and Development of sustainability and management, GDP-Gross Domestic Product, Significance of GDP.
2. Sustainability- concept, needs and challenges
3. Economic, social and aspects of sustainability from unsustainability to sustainability
4. Millennium development goals, and protocols
5. Sustainable Development Goals-targets, indicators and intervention areas
6. Principles of green chemistry, Climate change
7. Global, Regional and local environmental issues and possible solutions- case studies
8. Role of non-governmental organization, Concept of carbon credit, carbon footprint
9. Environmental management in industry - A case study.

Development: A Broader Perspective SESSION 28

It's a process that aims to create **progress and positive change** across all aspects of society, including:

- **Economic development:** Improving living standards and economic opportunities for all.
- **Environmental development:** Protecting and conserving the environment for future generations.
- **Social development:** Ensuring equal access to education, healthcare, and other social services.

Sustainable development is the key concept here. It means meeting the needs of the present without compromising the ability of future generations to meet their own needs.

Definition: -

Development is a continuous process that fosters **growth, progress, and positive change** in a country's economic, environmental, and social well-being, all while ensuring the **responsible use of environmental resources**.

1.National Development:

These are large-scale projects planned and implemented within a national framework. They aim to significantly contribute to a country's overall success and its standing in the international community. Examples might include infrastructure development projects or major economic reforms.

2.Major Developments:

These are specific projects that can have a significant local or regional impact. They can encompass a wide range of areas, such as:

- Fish farms
- Office buildings
- Storage and distribution centers
- Housing projects
- Renewable energy facilities
- Waste management facilities
- Mining operations

It's important to ensure that both national and major development projects are undertaken with a sustainable development approach, considering their economic, environmental, and social impact.

Types of Development

Development initiatives can be categorized according to their scope and impact:

- **National Developments:** These projects significantly contribute to a country's overall success, often involving large-scale infrastructure or economic initiatives.

- **Major Developments:** These projects aim to improve a country's economic power, such as building new industries or transportation networks.

- **Local Developments (Most Common):** These small-scale developments are the most frequent type and influence our daily lives. Examples include:

  - House extensions
  - Conversions of existing buildings
  - Small to medium-sized housing projects
  - Small-scale renewable energy installations (like solar panels on a house)

Principles of Development:

These seven principles are crucial for ensuring successful development projects:

1. **Equitable Use:** Benefits of the development should be distributed fairly and consider the needs of all stakeholders.
2. **Flexibility in Use:** The development should be adaptable to accommodate future changes or evolving needs.
3. **Simple and Intuitive Use:** The development should be easy to understand and navigate for users.
4. **Perceptible Information:** Information about the development should be clear, accessible, and readily understandable.
5. **Tolerance for Error:** The development should be designed to accommodate human error and minimize negative consequences.
6. Low physical effort.
7. Size and space for approach and use.

Factors Affecting Development

1. **Natural Resources:**
   - Examples: Trees, soil, water, minerals, coal, oil, etc.
   - Importance: Create jobs and generate wealth through sales.

2. **Power and Energy Resources:**
   - Examples: Oil, gas, coal, and water.
   - Importance: Provide power and energy for production within the country.
   - Note: These resources are natural and can be depleted quickly.

3. **Capital Accumulation:**
   - Importance: More capital allows for job creation.
   - Impact: Low capital countries may have lower living wages and higher unemployment.

Factors Affecting Development

4. **Technological Resources:**

   Examples: Computers, cell phones, etc.

   Importance: Enable advanced technologies, boosting business capabilities and economic development.

   Impact: Countries with limited technological resources experience slower economic development.

5. **Available Labor Force:**

   Importance: A skilled workforce drives development.

6. **Transportation and Communication Infrastructure**

7. **Education and Training**

Development: Advantages & Disadvantages

Employee Development:

Pros:

1. Happier & more motivated employees (increased job satisfaction)
2. More efficient processes (financial gain)
3. Adaptability to new technologies & methods

Economic Development:

Pros:

1. Increased state capacity (ability to provide public goods)
2. Overall economic growth

Disadvantages of Development (Broader Context):

1. Population growth
2. Weak governance & rapid urbanization (challenges managing growth)
3. Poverty (unequal distribution of benefits)
4. Pollution (environmental impact)

What is GDP?

- GDP stands for **Gross Domestic Product**.
- It measures the **total value of all final goods and services produced** within a country's borders in a given year.
- **Key Point:** It's the **broadest indicator** of a nation's economic activity.

Types of GDP

There are three main ways to measure GDP, each focusing on a different aspect:

1. **Nominal GDP:** This is the total value of goods and services produced at **current market prices**. It's useful for understanding the current size of the economy, but doesn't account for inflation.
2. **Real GDP:** This is the total value of goods and services produced adjusted for **inflation**. It allows for comparisons of economic growth over time by using constant prices (usually from a base year).
3. **Actual GDP (Not a standard term):** This term isn't commonly used by economists. It likely refers to an estimate of GDP for a shorter period than a year, like a quarter or month.

Importance of GDP

GDP is a crucial economic indicator because it tells us:

1. The **current state of a country's economy**.
2. How economies of different countries compare (**economic performance**).
3. Helps in **formulating economic policies**.
4. Provides a basis for understanding the **size and development** of an economy.

Calculating GDP

Economists use three main approaches to calculate GDP:

1. **Value Added Approach:** This method sums the value added at each stage of production to get the final value of the good or service.

2. **Income Approach:** This method totals the income earned by all factors of production (labor, capital, land, etc.) in the economy.

3. **Expenditure Approach (most common):** This method adds up the spending by all sectors of the economy: consumption, investment, government spending, and net exports (exports minus imports).

Expenditure Approach for GDP

Calculating GDP: What's Produced is What's Spent

The expenditure approach is a way to calculate a country's Gross Domestic Product (GDP) by looking at all the final spending on goods and services produced within the economy. In simpler terms, it measures the total value of everything bought and sold domestically over a specific period.

Here's the formula: **GDP (Y) = C + I + G + (X - M)**

- **Y:** Gross Domestic Product
- **C:** Consumption spending by households
- **I:** Gross private investment (businesses spending on things like equipment and buildings)
- **G:** Government spending on goods and services
- **X:** Exports (goods and services sold to other countries)
- **M:** Imports (goods and services bought from other countries)

Breaking it Down:

Imagine all the money spent in an economy flows into a giant bucket. This approach looks at where that money ultimately lands:

- **Consumers:** How much are people spending on groceries, clothes, entertainment, etc.?
- **Businesses:** How much are businesses investing in new equipment, factories, or software?
- **Government:** How much is the government spending on infrastructure, healthcare, education, etc.?
- **Net Exports:** The difference between what a country exports (selling goods and services to other countries) and what it imports (buying goods and services from other countries).

By adding up all this spending, we get a good picture of the total value of goods and services produced domestically.

Advantages of GDP (the Good)

1. **Easy to Understand:** GDP is a broad indicator of a country's economic activity.
2. **Growth Tracking:** It's easy to calculate growth in GDP as a percentage, allowing for comparisons over time.
3. **International Benchmarking:** Countries use the same formula, making it easier to compare GDP between nations.
4. **Cost-Effective:** Data collection for GDP is relatively cheap and efficient.
5. **Reliable Indicator:** The consistent formula across countries makes GDP a reliable benchmark.
6. **Policy Tool:** Governments use GDP to measure the success of economic policies.

But It's Not Perfect: Limitations of GDP (the Not-So-Good)

Disadvantages of GDP

1. GDP only considers transactions with a monetary value. It ignores important aspects like unpaid work (caring for children, housework) and bartering.
2. **Well-being Blind spot:** GDP growth doesn't directly translate to a better quality of life. It doesn't consider factors like happiness, healthcare, or education.
3. **Sustainability Concerns:** High GDP can mask environmental damage caused by economic activity. It doesn't show if growth can be sustained without harming the environment.
4. **Income Inequality:** GDP is a national total, not an individual measure. It doesn't show how wealth is distributed among citizens.
5. **Hidden Activities:** Informal sectors (black markets) and overseas income are often excluded from GDP calculations.
6. **Inflation & Manipulation:** High inflation can make GDP look bigger even though there's no real growth. Governments might manipulate figures for political reasons.
7. **Ethical Concerns:** GDP doesn't consider the ethics of production processes. For example, production of harmful goods can boost GDP.
8. **Past Focus:** High GDP in the past doesn't guarantee future prosperity. We need to focus on sustainable practices.

Sustainability

Sustainability is all about meeting our current needs without harming the ability of future generations to meet their own. In other words, it's about using resources wisely and protecting our planet for the long term.

Needs:

1. **Protects biodiversity:** Sustainable practices help conserve ecosystems and the incredible variety of life on Earth.
2. **Protects the environment:** Sustainability is all about protecting our natural environment, from clean air and water to forests and wildlife.
3. It is the choice of nontoxic materials
4. Sustainability protects biodiversity and also improves public health
5. It minimizes waste generation and reuses the resources
6. It is used for lifecycle analysis.

Concepts of sustainability

1. Developing appropriate technology
2. Reduce, Reuse and Recycle (3R approach)
3. Providing environmental education and awareness
4. Consumption of renewable resources
5. Conservation of nonrenewable resources
6. Population control

Significance of Sustainability:

1. Protects the environment for future generations.
2. Ensures a healthy planet for humans and other species.
3. Promotes economic development through efficient resource use.
4. Creates a more just and equitable society.

1.Economic Sustainability

- **Definition:** An organization's ability to manage resources and generate profits responsibly in the long term.

- **Examples:**
  1. **Unilever**: Implemented measures like recycling programs and promoting responsible consumption.
  2. **Suez**: Reduced emissions by using renewable energy and conserving natural habitats.

Economic and Social Challenges of Sustainability

1.Economic Sustainability

Challenges:

- High unemployment and poverty.
- Increasing inequality.
- Disruption of economic activities (e.g., tourism due to pandemics).
- Volatile growth dependent on a single source.
- Low productivity due to poor education and skill development.
- Skills mismatch between workforce skills and job demands.
- Lack of quality jobs.
- Macroeconomic instability and financial shocks.

2. Social Sustainability

Definition: Strengthening the cohesion and stability of social groups.

Examples:

1. CEMEX: Offers decent housing programs and loans to support communities.
2. Grupo Gigantic: Contributes resources to social causes like education and healthcare.

Challenges: (difficult to measure but significant)

1. Human rights violations.
2. Unfair labor practices.
3. Poor living conditions.
4. Health and safety risks.
5. Lack of well-being, diversity, and equity.
6. Work-life balance issues.
7. Limited empowerment opportunities.
8. Weak community engagement.

Aspects of Sustainability

There are four interconnected aspects of sustainability:

1. **Environmental:** Maintaining the natural world (ecosystems) for all living things.
2. **Social:** Ensuring equity within and between generations and social groups.
3. **Cultural:** Respecting diverse worldviews and enabling participation in decision-making.
4. **Economic:** Using resources efficiently and creating a fair trading system while considering environmental limitations.

From Unsustainability to Sustainability

Unsustainability

•Our ancestors used resources wisely for their needs, not greed.

•We are exploiting limited resources for our wants, not needs.

Characteristics of Unsustainability

1. We forget our responsibility towards the environment.
2. We degrade available resources.
3. We don't care about the needs of future generations.
4. Unsystematic planning damages natural and human-made resources.
5. It leads to extreme environmental degradation and harm to living organisms.

From Unsustainability to Sustainability

Causes of Unsustainability

1. Developing countries contribute to degradation, but developed countries pollute more.
2. Rising population puts pressure on resources.
3. We see the environment as ours only, neglecting other living things.
4. We prioritize luxury lifestyles over animal habitats.
5. We overuse resources from the Earth.
6. We create harmful chemicals that pollute the environment.
7. Deforestation (cutting trees faster than they grow) causes global warming.
8. Using plastic bags increases pollution.

Characteristics of Sustainability

1. Reduces greenhouse gas emissions to fight global warming and protect the environment.
2. Uses natural and biodegradable materials to minimize environmental impact.
3. Emphasizes renewable energy sources like wind and solar power.
4. Follows non-polluting construction practices.
5. Protects natural habitats.
6. Improves human quality of life.
7. Minimizes depletion of natural resources.
8. Teaches respect and care for all living things.
9. Ensures future generations can meet their needs.

Differences Between Sustainability and Unsustainability

S.No.	***Sustainability***	***Unsustainability***
1	Prevention of natural resources	Exploitation of natural resources is faster than the planet can handle and replenish.
2	LoLong-term economic growth that minimizes negative impacts on the environment and society.	Not quantifying ecosystem services and increased vulnerability to crises.
3	Promotes equality, diversity, social cohesion, and democracy.	Labour laws, human rights, gender inequality

Millennium Development Goals (MDGs) Session 31

8 International Development Goals:

1. Eradicate Extreme Poverty and Hunger
2. Achieve Universal Primary Education
3. Promote Gender Equality and Empower Women
4. Reduce Child Mortality
5. Improve Maternal Health
6. Combat HIV/AIDS, Malaria, and Other Diseases
7. Ensure Environmental Sustainability
8. Develop a Global Partnership for Development

Sustainability Protocols and Sustainable Development

Sustainability Protocols

1. Define sustainability standards and certifications.
2. Voluntary guidelines for producers, retailers, etc. to show commitment to environmental, social, and ethical practices.
3. Over 400 standards exist globally.

Benefits:

1. Raise awareness and policy support.
2. Create clear guidelines and goals.
3. Maintain consistency in developing a green economy.
4. Third-party verification ensures adherence to standards.

Eco-Labels and Standards for Food Products

1. Triple bottom line approach: considers environmental, social, and economic factors.
2. Defines practices for sustainable crop growing and ethical resource harvesting.
3. Examples: responsible fishing, fair wages for farmers.
4. Verification process ensures compliance with standards and traceability throughout the supply chain.
5. Often results in consumer-facing labels.

Goals of True Sustainable Development

- **Optimum resource use:** Minimize waste, pollution, and byproducts while maximizing productivity.
- **Equity:**
  1. **Intergenerational:** Leave a healthy and resourceful environment for future generations.
  2. **Intragenerational**: Rich countries' technology should support poor countries' economic growth and reduce wealth gaps.

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Principles of Sustainable Development

1. **Pollution Prevention and Waste Reduction:** Emphasize recycling, reuse, and creating longer-lasting products.
2. **Renewable Energy Sources:** Rely on sun, wind, water, and biomass for energy
3. **Biodiversity Conservation:** Protect and sustain Earth's biodiversity.
4. **Sustainable Practices:** Discourage activities that degrade the Earth.
5. **Poverty Reduction and Population Control:** Reduce poverty and population growth rates.
6. **Energy Efficiency:** Avoid using high-quality energy for tasks not requiring it.

UN Sustainable Development Goals (SDGs)

There are 17 SDGs, including:

1. No poverty (SDG 1):
2. Zero hunger, achieve food security and improved nutrition (SDG 2):
3. Ensuring Good health and well-being of all age grou (SDG 3):
4. Ensuring Quality education (SDG 4):
5. Gender equality (SDG 5):
6. Clean water and sanitation (SDG 6):
7. Affordable, reliable and clean energy for all (SDG 7):
8. Promoting decent work and economic growth (SDG 8):
9. Building resilient infrastructure, sustainable Industrialization and innovation (SDG 9)
10. Reduced inequalities within and among countries (SDG 10)

UN Sustainable Development Goals (SDGs)

11. Making Sustainable cities and communities (SDG 11):
12. Ensuring sustainable and responsible consumption and production (SDG 12):
13. Taking urgent action to combat Climate change and its impacts (SDG 13):
14. Conserving Life below Ocean, sea and Marine resources (SDG 14):
15. Protecting, restoring and promoting the sustainable use of Life on land (SDG 15)
16. Promoting Peace, justice, and strong institutions (SDG 16):
17. Revitalising the global partnerships for sustainable development (SDG 17):

The Sustainable Development Goals aim to eradicate extreme poverty, improve food security and nutrition, reduce maternal mortality, and address health and well-being issues. These goals include eradicating extreme poverty, ensuring equal access to resources, building resilience against climate-related disasters, promoting sustainable food production, and ensuring functioning food commodity markets.

The Sustainable Development targets

1. Eradicate extreme poverty for all the people everywhere.

2. Reduce at least by half the proportion of men, women and children of all ages living in poverty.

3. Implement nationally appropriate social protection systems and measures for all floors.

4. Ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, natural resources and new technology and financial services.

5. Built the resilience of the poor and those in vulnerable situations and reduce their exposure to climate-related events and disasters.

6. Ensure significant mobilization of resources from a variety of sources from developed countries to less developed countries.

The Sustainable Development targets

7. End hunger and ensure access by all people, in particular the poor people and people in vulnerable situations.

8. End all forms of malnutrition.

9. Double the agricultural productivity and income of small scale food producers.

10. Ensure sustainable food production systems and implement resilient agricultural practices.

11. Maintain the genetic diversity of seeds, cultivated plants and farmed animals.

12. Increase investment in rural infrastructure. agricultural research and technological development to enhance agricultural productive capacity.

13. Correct and prevent trade restrictions and distortions in world agricultural markets.

14. Adopt measures to ensure the proper functioning of food commodity markets.

The Sustainable Development targets

15. Reduce the global maternal mortality ratio to less than 70 per 1,00,000 live births.

16. End preventable deaths of newborns and children under 5 years of age.

17. End the epidemics of AIDS, tuberculosis, malaria, hepatitis, and water-borne diseases.

18. Reduce by one-third premature mortality from non-communicable diseases through prevention and treatment.

19. Strengthen the prevention and treatment of substance abuse like narcotic drug abuse and harmful use of alcohol.

21. Ensure universal access to sexual and reproductive healthcare services like family planning.

22. Achieve universal health coverage including financial risk protection.

23. Reduce the number of deaths and illnesses from hazardous chemicals and air, water, and soil pollution.

Sustainable development indicators

1. Proportion of the population living below the international poverty line by sex, age, employment status.
2. Proportion of population covered by social protection floors.
3. Proportion of the total adult population with secure tenure rights to land with legally recognized documentation.
4. Proportion of local governments that adopt and implement local disaster risk reduction strategies in line with national disaster risk reduction strategies.
5. Proportion of total government spending on essential services.
6. Prevalence of undernourishment.

Sustainable development indicators

8. Prevalence of stunting and malnutrition among children and 5 years of age and prevalence of anemia in women aged 15 to 49 years.
9. Average income of small-scale food producers.
10. Proportion of agricultural area under productive and sustainable agricultural practices.
11. Proportion of local breeds classified as being at risk of extinction.
12. Total official flows to the agricultural sector.
13. Agricultural export subsidies.
14. Indicator of food price anomalies.
15. Maternal mortality ratio.
16. Under 5 mortality rate.

Sustainable development indicators

17. Hepatitis B (or) Tuberculosis incidence per 100,000 population. Malaria incidence (or) HIV infections per 1000 population.
18. Suicide mortality rate.
19. Alcohol per capita consumption within a calendar year in liters of pure alcohol.
20. Death rate due to road traffic injuries.
21. Proportion of women of reproductive age, who need family planning.
22. Coverage of essential health services.
23. Mortality rate due to unsafe water, sanitation and lack of hygiene.

Intervention areas of Sustainable development goals

Implementation of sustainable development goals are somewhat difficult because of the following interventions.

1. Climate change.
2. Use of natural resources.
3. Waste production.
4. Water pollution.
5. Deforestation.
6. Overfishing
7. Poverty
8. Ocean acidification
9. Air pollution

GREEN CHEMISTRY

Session 33

Green chemistry is the chemistry that involves designing and production of chemicals without polluting the environment. ie., Green chemistry is the pollution free chemistry.

12 Principles of Geen Chemistry

1. Prevention of wastes

It is better to prevent wastes than to treat or clean up wastes after they have been created.

2. Atom economy

Synthetic methods should be designed to maximize the incorporation of all materials used in the process of formation of final product.

3. Less hazardous chemical synthesis

Synthetic methods should be designed to use and generate materials, which possess no toxicity or little to human health and the environment.

4. Designing safer chemicals

Chemical substances should be designed to affect their desired functions during minimizing their toxicity.

5. Safer solvents and auxiliaries

The use of auxiliary substances (eg., toxic solvents, separation agents, etc.,) should be made unnecessary wherever possible and not harmful when used.

6. Design for energy efficiency

Energy requirements, during the chemical processes, should be minimized. If possible, synthetic methods should be carried out at ambient temperature and pressure.

7. Use of renewable feedstock

Raw materials or feed stocks used should be renewable rather than depleting.

8. Reduce derivatives

Unnecessary derivatives should be minimized or avoided if possible, because such steps require additional reagents and generate wastes.

GREEN CHEMISTRY

9. Catalysis

Catalytic reagents are superior to stoichiometric reagents.

10. Design for degradation

Chemical products should be designed in such a way that at the end of their function they break down into harmless degradation products and do not persist in the environment.

11. Real time analysis for pollution prevention

Analytical methodologies need to be developed to allow for real-time, in-process monitoring and control prior to the formation of hazardous substances.

12. Inherently safer chemistry for Accident prevention

Substances and some form of a substance, used in a chemical process, should be chosen to minimize the chemical accidents, including releases, explosions and fires.

Climate change

What is climate change?

- Climate change is a change in the statistical distribution of weather patterns when that change lasts for an extended period of time (i.e., decades to millions of years).
- Climate change is the rise in average surface temperatures on Earth, mostly due to the burning of fossil fuels.
- Climate change, also called global warming, refers to the rise in average surface temperatures on Earth.

Climate Change (Causes, Effects, and Solutions):

- Caused by greenhouse gases and ozone depletion.
- Leads to changes in temperature, weather patterns, rainfall distribution, and wind/ocean currents.
- Effects include disrupted agriculture, floods, droughts, rising sea levels, harm to ecosystems, and increased heat-related illnesses.

Solutions involve reducing greenhouse gas emissions through:

1. Burning less fossil fuels.
2. Using renewable energy sources (solar, geothermal).
3. Reducing deforestation and planting trees.
4. Improving energy efficiency.

Climate Change Solutions

1.Reduce Reliance on Fossil Fuels:

2.Transportation: - Reduce the use of petrol and diesel vehicles, planes, and ships.

- Switch to electric vehicles.

3.Energy Efficiency at Home:

4.Dietary Changes:

- Reduce meat and dairy consumption. Consider vegetarianism.

5.Natural Solutions:

- Plant trees in suitable locations to absorb carbon emissions. Protect existing forests.

6.Reduce Consumption:

7.Global Action:

- Reduce carbon pollution to mitigate climate change impacts.
- Provide financial support to developing countries for adaptation.

Climate Change Impact on Chennai: Case Study Session 34

Location: East Coast Road (ECR) and Old Mahabalipuram Road (OMR)

Key Points: ECR and OMR are highly vulnerable to climate change due to Chennai's:

- Land-use patterns (increased construction)
- Growing population
- Resource overuse

Climate impacts becoming more frequent and severe:

- Droughts
- Floods
- Heavy rainfall
- Strong winds

Chennai experiences both water scarcity and flooding:

- Summers: Water shortages
- Monsoons: Devastating floods

Climate Change Impact on Chennai: Case Study

Reasons for Flooding:

- Altered natural water flow:
  - Increased buildings and infrastructure block natural drainage channels.
  - Water cannot reach the sea via regional watersheds.

Remedies:

- Researchers predict Chennai's climate will be:
  - Unpredictable
  - More Aggressive
- Reducing greenhouse gas emissions is crucial to combat climate change.

Climate Change Impacts in Chennai, Ennore: Summary

Industrial Pollution:

- Six major industries in Ennore (including Barve thermal power station) violate air pollution norms for nearly 60% of the year.
- These industries release massive amounts of pollutants like particulate matter, SO2, and NO2, exceeding the area's carrying capacity.

Health Impacts:

- Increased respiratory and gynecological problems reported in Ennore, particularly affecting children and women.

Environmental Damage:

- Fly ash from power plants disrupts natural drainage patterns and harms local ecology.
- Hot water discharge from power plants disrupts marine life, potentially creating "marine deserts."

Overall Impact:

Climate change intensifies these issues, leading to the potential for increased flooding, droughts, and ocean desertification.

Role of Non Governmental Organizations NGOs

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1. To protect environment resources and their economical use.
2. To organise courses like lectures, seminars.
3. To create informational materials like newsletters, brochures, booklets, articles and videos.
4. To analyse and monitor the resources independently.
5. To protect the rights of the citizens to a healthy environment and the consumer's rights to clean environment.
6. To keep vigil in the surrounding area, well, river, lake, land and air against pollution and reporting to state board/central board.

7. To conduct sampling and analysis of well/river water to establish its quality.
8. To publish the notified restricted areas where industries, industrial operations should not be carried out.
9. To provide information on fish killing to the environment not noticed by the state board.

Some important NGOs in India

1. Centre for Science and Environment (CSE), New Delhi.
2. Centre for Environmental Education (CEE), Ahmadabad.
3. Environmental Society of India, Chandigarh.
4. Madras Environmental Society, Chennai.

Carbon Credit

Definition and Concept of Carbon Credit:

- Carbon credits represent a unit of measurement for greenhouse gas (GHG) emissions. One carbon credit is equivalent to the removal or reduction of one metric ton of carbon dioxide (CO2) or its equivalent in other GHGs.
- The concept revolves around incentivizing companies or entities to reduce their emissions below a certain baseline by providing them with tradable credits, thus creating a market-based mechanism to mitigate climate change.

Types of Carbon Credits

1. **Certified Emission Reductions (CERs):** Generated under the Clean Development Mechanism (CDM) of the Kyoto Protocol. Projects in developing countries can earn CERs by reducing emissions or removing carbon from the atmosphere.

2. **Emission Reduction Units (ERUs):** Generated under Joint Implementation (JI) projects, allowing developed countries to earn credits for emission reduction projects in other developed countries.

3. **Verified Emission Reductions (VERs):** Also known as voluntary carbon credits, these are generated from projects that are not bound by international agreements but voluntarily reduce emissions.

How to Acquire Carbon Credits

1. **Identify Emission Reduction Opportunities:** Assess operations to identify areas where emissions can be reduced through efficiency improvements, renewable energy adoption, or carbon capture technologies.
2. **Implement Emission Reduction Projects:** Execute projects that lead to measurable emissions reductions or removals, such as renewable energy installations, reforestation efforts, or energy efficiency upgrades.
3. **Certification and Verification:** Engage with accredited certifiers to verify emission reductions according to established standards such as the Clean Development Mechanism (CDM) or Verified Carbon Standard (VCS).
4. **Registration and Issuance:** Register projects with relevant authorities and obtain carbon credits upon successful verification, which can be traded on carbon markets.

Advantages of Carbon Credits

1.Incentivize Emission Reduction: Provide financial incentives for companies to invest in emission reduction projects, fostering innovation and technological advancements in clean energy and sustainability.

2.Cost-Effective Climate Mitigation: Offer a cost-effective way to achieve emission reduction targets by allowing companies to invest in projects with the greatest potential for reducing emissions.

3.Promote Sustainable Development: Support sustainable development goals by directing investment towards renewable energy, energy efficiency, and environmental conservation projects, particularly in developing countries.

Disadvantages of Carbon Credits

1.Risk of Greenwashing: Some projects may not deliver the anticipated emissions reductions, leading to accusations of greenwashing or misleading claims about environmental impact.

2.Complexity and Administrative Burden: The process of acquiring and managing carbon credits can be complex and costly, particularly for smaller organizations without dedicated resources or expertise.

3.Volatility and Speculation: Carbon markets can be subject to volatility and speculation, leading to fluctuating prices and uncertain returns on investment, which may deter long-term commitments to emission reduction projects.

Carbon Footprint

•A measure of the total amount of greenhouse gases, primarily carbon dioxide, emitted directly or indirectly by human activities.

Individual Carbon Footprint

- Refers to the amount of greenhouse gas emissions produced by an individual's activities, including transportation, energy consumption, and lifestyle choices.

Sources of Carbon Footprint

1. **Transportation:** Emissions from vehicles, including cars, airplanes, and public transportation.
2. **Energy Use:** Emissions from electricity and heating/cooling systems in homes and buildings.
3. **Food Production:** Emissions associated with the production, transportation, and packaging of food.
4. **Waste Management:** Emissions from landfills and waste incineration.
5. **Consumption Habits:** Emissions from the purchase and use of goods and services.

Causes of High Carbon Footprint

1.Dependency on Fossil Fuels: Reliance on coal, oil, and natural gas for energy.

2.High Consumption Lifestyle: Overconsumption of goods and resources.

3.Inefficient Transportation: Use of personal vehicles instead of public transportation or alternative modes of transportation.

4.Wasteful Practices: Excessive use of disposable products and inefficient use of energy.

How to Lower Carbon Footprint - 15 Ways

1. **Reduce Energy Consumption:** Use energy-efficient appliances and lighting, and turn off electronics when not in use.
2. **Switch to Renewable Energy:** Install solar panels or purchase renewable energy from utilities.
3. **Drive Less:** Walk, bike, carpool, or use public transportation whenever possible.
4. **Choose Sustainable Transportation:** Opt for electric or hybrid vehicles, or use alternative fuels like biofuels.
5. **Reduce Air Travel:** Minimize air travel or choose more fuel-efficient airlines when flying.
6. **Eat a Plant-Based Diet:** Reduce meat consumption and choose locally sourced, sustainable foods.
7. **Reduce Food Waste:** Plan meals, store food properly, and compost food scraps.
8. **Support Sustainable Products:** Choose products with minimal packaging and environmentally friendly certifications.

How to Lower Carbon Footprint - 15 Ways

9. **Conserve Water:** Fix leaks, use water-saving appliances, and reduce water consumption.
10. **Reuse and Recycle:** Minimize waste by reusing items and recycling materials whenever possible.
11. **Plant Trees:** Support reforestation efforts and plant trees to absorb carbon dioxide.
12. **Green Your Home:** Improve insulation, use eco-friendly materials, and invest in energy-saving upgrades.
13. **Support Climate Policies:** Advocate for policies that promote renewable energy and carbon reduction.
14. **Educate Others:** Raise awareness about climate change and encourage others to take action.
15. **Offset Emissions:** Support carbon offset projects that reduce greenhouse gas emissions, such as forest conservation or renewable energy initiatives.

Environmental Management (EM) Session 36

•Set of practices that help organizations (public or private) to reduce their environmental impact and improve efficiency.

Goals of EM

1. **Minimize harm:** Reduce negative effects on various environmental components (air, water, soil, etc.) caused by human activities.
2. **Conserve resources:** Protect and manage environmental resources for future generations.
3. **Enhance value:** Improve the overall health and quality of environmental components.

Key Functions of EM

1.Monitoring: Track environmental performance to identify areas for improvement.

2.Evaluation: Analyze data to assess the effectiveness of environmental management plans.

3.Reorientation: Adapt and adjust plans based on findings from monitoring and evaluation.

4.Protective Measures: Implement strategies to prevent environmental damage during project construction and operation.

5.Human Well-being: Improve the quality of life for people by protecting the environment.

Broader Benefits of EM

1. **Problem-solving:** Proactive approach to preventing and addressing environmental issues.
2. **Regulations:** Comply with environmental regulations to avoid fines and penalties.
3. **Sustainability:** Develop strategies for long-term, sustainable development that considers both environmental and economic needs.
4. **Technology:** Identify and adopt eco-friendly technologies to minimize environmental impact.
5. **Business Protection:** Mitigate risks associated with environmental damage for businesses.

Principles of Environmental Management

7 Guiding Principles for Sustainable Decisions

Environmental management is all about making choices that benefit both the environment and society. Here are 7 key principles that guide these decisions:

1. **Polluter Pays Principle (PPP)**
2. **Precautionary Principle (PP)**
3. **Sustainable Use Principle (SUP)**
4. **Responsibility Principle (R)**
5. **Participation Principle (P)**
6. **Prevention Principle (P)**
7. **Proportionality Principle (P)**

These principles encourage polluters to adopt cleaner practices, prioritize safety, use resources wisely and efficiently, share responsibility for the environment, promote public participation, prevent pollution, and balance the environmental costs and benefits of development projects.

Environmental Management: 5 Key Steps

1. Environmental Policy:

- **Define Your Commitment:** Develop a clear policy outlining your organization's dedication to environmental responsibility and sustainability.
- **Set Goals & Targets:** Establish specific, measurable objectives to track your environmental performance.
- **Integrate with Strategy:** Make environmental considerations a core part of your overall strategic planning.

2. Planning:

- **Identify Impacts:** Assess the environmental footprint of your activities, products, and services.
- **Know the Rules:** Understand legal and regulatory requirements related to environmental management.
- **Develop a Plan:** Create procedures to reduce or prevent environmental harm.
- **Assign Resources:** Allocate necessary resources and assign roles to support environmental goals.

Environmental Management: 5 Key Steps

3. Implementation:

- **Put it into Action:** Execute the plans and procedures developed during the planning phase.
- **Educate & Empower:** Train employees on their environmental responsibilities.
- **Integrate Everyday:** Consider environmental factors in daily operations and decisions.
- **Minimize the Impact:** Implement technologies and practices to reduce environmental footprint.

4. Checking and Correction:

- **Track Your Progress:** Monitor key performance indicators (KPIs) to measure progress towards environmental goals.
- **Regular Checkups:** Conduct inspections and audits to identify areas for improvement and non-compliance.
- **Learn from Incidents:** Investigate environmental issues like accidents and complaints.
- **Fix and Prevent:** Take corrective and preventive actions to address problems and stop them from happening again.

Environmental Management: 5 Key Steps

5. Management Review:

- **Leadership Evaluation:** Top management periodically reviews the environmental management system (EMS).
- **Assess Effectiveness:** Evaluate how well your policies, plans, and procedures achieve environmental goals.
- **Stay Compliant:** Ensure compliance with legal requirements and other environmental standards.
- **Continual Improvement:** Identify opportunities to constantly improve your environmental performance and innovate.

Environmental Management Characteristics & Benefits

Environmental Management plays a crucial role in today's world. Here's a quick breakdown of its key characteristics and the benefits it brings:

Characteristics:

1. **Supports Sustainable Development:** Environmental management aims to meet our present needs without compromising the ability of future generations to meet theirs.
2. **Multidisciplinary Approach:** It requires expertise from various fields like ecology, engineering, economics, and law.
3. **Integrates Different Viewpoints:** Environmental decisions consider social, economic, and ecological factors.
4. **Natural & Social Sciences:** Bridges the gap between scientific understanding and social impacts.
5. **Scalable - Local to Global:** Environmental issues can be local (pollution) or global (climate change), requiring management at different scales.
6. **Human-Centered Focus:** Recognizes the human impact on the environment and seeks solutions

Benefits of Environmental Management:

1.Improved Environmental Performance: Measures and minimizes negative environmental impacts.

2.Enhanced Compliance: Ensures adherence to environmental regulations and avoids penalties.

3.Pollution Prevention: Reduces pollution at the source, leading to a cleaner environment.

4.Resource Conservation: Promotes efficient use of resources like water and energy.

5.Attracting New Customers/Markets: Environmentally conscious practices can attract customers who value sustainability.

6.Problem-Solving & Efficiency: Identifies and solves environmental issues, leading to cost reductions.

7.Employee Morale: Fosters a sense of responsibility and pride in working for an environmentally conscious company.

Positive Public Image: Improves reputation with stakeholders like regulators, lenders, and the public

Case Studies

Electronic Waste (E-waste) Recycling: Dell's Legacy of Good Program

- **Problem:** Electronic waste (e-waste) is a growing environmental concern.
- **Solution:** Dell's "Legacy of Good" program aims to reduce e-waste and create more sustainable products.

Program Goals:

- Use 50 million pounds of recycled plastic and other sustainable materials.
- Design 100% recyclable or compostable packaging.
- Recover 2 billion pounds of electronic waste.

Actions Taken:

- Used 4.5 million kilos of recycled plastic in monitors and desktops.
- Eliminated 20 million pounds of packaging waste, saving $18 million.
- Replaced non-biodegradable packaging with bamboo and mushroom-based alternatives.

Importance: Encourages others to see waste as a valuable resource.

Key Takeaway: Cleaner production, a proactive approach to minimizing waste generation at the source, is the most effective environmental management strategy.

Biomedical Waste Management in Nepal

- **Problem:** Improper management of medical waste poses a health and environmental threat in Nepal.
- **Solution:** Develop and implement proper waste management practices.

Challenges:

- Lack of proper disposal methods (incineration, open burning, or dumping)
- Limited training for medical professionals

Initiatives:

- National healthcare waste management guidelines developed by Nepal Health Resource Council and WHO.
- Training manuals for medical professionals.

Importance: Proper management protects human health and the environment.

Municipal Solid Waste Management in Solapur City, India

Waste Generation:

Solapur City generates **large amount of Municipal Solid Waste (MSW)** per day

•Breakdown of waste:

1. Biodegradable: 50%
2. Recyclable: 25%
3. Green waste: 15.3%
4. Debris and silt: 9.9%

Challenges:

- Only 51% of the total waste is collected.
- Open dumping at landfill sites (Tuljapur road and Bhogaon) leads to:
    1. Contamination of soil and water
    2. Unpleasant odors
    3. Slow natural degradation

Municipal Solid Waste Management in Solapur City, India

Future Plans:

•An anaerobic digestion treatment plant is under construction.

Key Points:

1. Solapur faces challenges in managing its municipal solid waste.
2. Most of the waste is not collected and ends up in landfills.
3. Improper disposal methods cause environmental and health problems.
4. A new treatment plant offers hope for a more sustainable waste management system.